MOIRAIS: A Multi-Domain Scientific Computing Toolkit for Observational Inference, with Sociolegal, Signal-Processing, Cryptographic, and Spatial-Statistics Modules

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2026-05-09

Abstract

This study describes *MOIRAIS* (Methods for Observational Inference and Robust Analysis of Interventions in Scientific Experimentation), a dual-language (Python and R) open-source scientific-computing toolkit released alongside this paper. MOIRAIS is multi-domain by construction: a single package surface supports general-purpose causal inference and survey-weighted estimation, signal processing and spectral analysis (with applications to forensic audio and biomedical signals), classical and modern cryptographic primitives, spatial statistics, statistical-physics models of crime, classical-test-theory and item-response-theory psychometrics, automated scrapers for Canadian carceral and police- oversight corpora, and a primary implementation of the MRM (McNamara-Ruhela-Medina) framework for sociolegal data analysis [6]. The package ships 41 built-in datasets in a portable SQLite layer, a 36,476-file individual- function namespace, a 10-screen Textual terminal user interface, a multi-provider local-LLM chain with a vendored TurboQuant KV-cache compression [14], and a pure-Python emissions tracker covering 213 countries. The current Python test suite contains an extensive baseline test suite plus 55 tests covering the federal/provincial Mandela cross-jurisdiction modules, all passing under continuous integration; the Sphinx documentation site builds without warnings under strict mode.

1 Introduction

Scientific computing tooling is highly fragmented: causal inference, survey statistics, spatial analysis, signal processing, cryptography, and statistical physics each have their own ecosystems with distinct interfaces, data conventions, and packaging norms. Researchers who work *across* these domains — a population that includes anyone studying carceral / criminological data (which mixes survey design, causal estimands, spatial statistics, and self-exciting point processes) — pay a recurring tax in glue code, format conversions, and re-implementation of standard diagnostics that already exist in some other package.

MOIRAIS addresses this gap by providing a single namespace, a single dataset registry, and a single set of result containers (`RichResult`) that span all of the above domains. It is deliberately broader than its named primary application (the MRM framework on Canadian sociolegal data) and is intended to be useful for anyone working at the intersections of causal inference, signal processing, cryptography, and spatial / statistical-physics modelling.

The package is open source under the GNU General Public License v2 (Linus copyleft, deliberately chosen over GPL-3.0 for compatibility with the broader Linux-kernel-style ecosystem), and is released via PyPI, CRAN, and r-universe simultaneously. The R package mirrors the Python package function-for-function via the `moirais` R package whose source lives at `r-package/moirais/` in the same repository.

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This paper documents the package's scope and architecture; the mathematical foundations of the MRM primary application are developed in detail in the companion paper [6].

2 Architecture

2.1 Repository layout

The MOIRAIS source repository follows the standard scientific-Python layout. The Python package lives at `src/moirais/` (flat module layout, $\approx$ 60 top-level modules); the R package lives at `r-package/moirais/` (standard CRAN layout with `R/`, `man/`, `tests/`, `inst/`, `DESCRIPTION`, `NAMESPACE`). Documentation source sits at `docs/source/`, the test suite at `tests/`, and the continuous-integration workflows in `.github/workflows/`. The Sphinx site is built from `docs/source/` and published to GitHub Pages on each push to `main`.

2.2 The `moirais.fn` namespace

A central architectural feature is the `moirais.fn` namespace, which contains tens of thousands of individual function files (one function per file, with shared infrastructure in `fn/_containers.py`, `fn/_richresult.py`, `fn/_registry.py`, etc.). Each file is named under a short stable-identifier convention (typically 3–7 characters), e.g. `ate.py`, `difmh.py`, `kmo.py`, `huberw.py`. The intent is canonical short names per estimator/test/kernel/weight matrix; the registry at `fn/_registry.py` maps short names to fully-qualified callable paths and a one-line description used by `moirais.fn.cheatsheet(name)`. Older monolithic modules (`causal.py`, `psymet.py`, ...) re-export from `fn/` for backward compatibility, so user-facing code is unaffected by the per-file naming.

2.3 The `RichResult` container

Every analysis function in MOIRAIS returns a `RichResult` object — an `R`-style multi-section result container modelled on `psych::omega` output. A `RichResult` has structured fields for a title, summary lines, tables, warnings, an interpretation paragraph, and a typed payload, and inherits from `dict` so that `isinstance(result, dict)` returns `True` for any downstream code that expects raw dictionary output. The container is rendered as a multi-section paragraph block by default, and its payload is JSON-serialisable for archival.

2.4 Result containers and the `describe()` convention

In addition to the result, every callable that emits a `RichResult` ships a sibling `describe_<short>.md` documentation file that explains the function in pedagogical multi-section paragraphs (theory; assumptions; equations; example output; references). The `moirais.describe(callable)` helper renders these files inline at the REPL or in the TUI, so the user can ask, mid-session, "what is this function actually computing?"

2.5 Terminal user interface

MOIRAIS is terminal-first by design. A 10-screen TUI built on Textual provides interactive data exploration (Datasets), pipeline execution (Pipelines), diagnostics (Doctor), help and reference (Help), debugging (Debug), $\approx$ 900 statistical commands (Stat), a polyglot REPL (§8), settings, and an LLM chat (§9). The TUI requires no browser and runs on a remote ssh session; it is the recommended interface for users in secure or air-gapped environments.

3 Data infrastructure

3.1 The 41 built-in datasets

MOIRAIS ships 41 datasets in a portable `moirais_datasets.db` SQLite file (≈ 500 MB, tracked via Git LFS) that is bundled inside the Python package's `data/` directory and accessible to both Python (via the `sqlite3` stdlib module) and R (via `DBI + RSQLite`). The dataset catalogue covers Canadian public-health surveillance (Statistics Canada CPADS, CCS, CSADS, CSUS PUMF microdata; Health Infobase aggregate trend tables; CIHI hospital / health-system indicators; the 809,000-row indicator library), the open Ontario Tracking Information System (OTIS) provincial restrictive-confinement dataset, the Ontario Special Investigations Unit canonical 65-column case CSV produced by MOIRAIS's own scraper (§7), the nine Toronto Police Service open-data crime categories, and the Statistics Canada Crime Severity Index weights. An optional `download-bootstrap` subcommand fetches an additional 1.5 GB of bootstrap weights on demand.

3.2 Dataset-agnostic estimator interfaces

A design rule across the package is that no estimator hard-codes a column name. Every function takes column names as keyword arguments with sensible defaults, so the same callable runs unchanged across CPADS, OTIS, TPS, or any user-supplied `pandas.DataFrame`. The CPADS column contract is provided as a canonical reference (`moirais.cpads`) but is opt-in, not enforced.

3.3 The `DatasetRegistry`

A central `DatasetRegistry` class maps each dataset's short identifier (e.g. `otis-2025`, `cpads-2122`, `tps-assault`) to a loader function, a cached `ColumnProfile` per column, and a suggested-role heuristic that flags candidate `treatment / outcome / covariate / weight / id / stratum / cluster` columns. The registry powers the TUI's Datasets screen and the `moirais list-datasets` CLI.

4 Causal inference and survey statistics

The `causal`, `effects`, `investigation`, `ebac`, `survey`, and `inference` modules implement a comprehensive suite of causal estimators with full survey-weight support:

- Inverse-probability-weighting and Hájek-stabilised IPW (ATE, ATT, ATC).
- Augmented IPW (doubly robust), with the standard semiparametric efficiency bound when both nuisance models are correctly specified.
- Double / debiased machine learning (DML) with the partially linear (PLR) and interactive (IRM) score functions, via the `DoubleML` dependency [2].
- Propensity-score matching and Rosenbaum–Rubin subclassification.
- AIPW with a SuperLearner ensemble [13].
- Instrumental variables (2SLS / partially linear IV).
- Regression discontinuity (sharp and fuzzy).
- Difference-in-differences, including the Bacon decomposition.
- Sensitivity analysis: Rosenbaum bounds, the E-value, and a generic `verify_chi2()` helper for replication auditing.

The survey-weighted machinery uses a `SurveyDesign` class that parallels the R `survey` package [5] and supports stratification, clustering, probability weights, and calibration / bootstrap variance estimation. The Horvitz–Thompson and Hájek estimators are exposed as primitives for any user wishing to roll their own design.

5 Signal processing, cryptography, and adjacent modules

5.1 Signal processing and spectral analysis

The `signal_processing` and `homomorphic_deconvolution` modules expose a working stack for forensic-audio, biomedical-signal, and seismological applications: classical FFT and short-time Fourier transform, multi-taper spectral estimation, wavelet decomposition (continuous and discrete), cepstral methods, spectral subtraction, blind-deconvolution by cepstral liftering and homomorphic filtering, and time–frequency reassignment. The homomorphic-deconvolution path is exposed as a separate module because it has independent applications outside signal processing (notably in genomics).

5.2 Cryptography

The `crypto` module collects classical and modern cryptographic primitives suitable for teaching and for low-stakes research workflows: stream ciphers, block ciphers (DES, AES, 3DES), hash families (MD5, SHA-2, SHA-3, BLAKE2/3), HMAC, classical substitution and transposition ciphers (Caesar, Vigenère, Hill, playfair), elliptic-curve primitives (X25519, Ed25519), and constant-time comparison helpers. The module is not intended as a substitute for production-grade libraries but as a teaching layer that shares result containers with the rest of MOIRAIS.

5.3 Spatial statistics

The `spatial` module exposes Moran's I (global, local, bivariate), Ripley's K and L, the Getis–Ord G^* statistic, kernel density estimation, DBSCAN clustering on geo-coordinates, and a polygon-based Moran's I implementation for neighbourhood-aggregate data (used heavily by the TPS- aggregated workloads in §6).

5.4 Statistical physics of crime

A `tps_statphysics` module implements the Short–Brantingham–D'Orsogna reaction–diffusion crime model [10], the Lévy-flight tail-index estimator (Hill estimator on inter-event spatial displacements), the Bettencourt–Lobo–West urban-scaling exponent [1] fit by HC3-robust OLS, and a Lotka–Volterra fit for paired police–crime annual time series. A separate `tps_hawkes_advanced` module fits non-stationary Hawkes self-exciting processes with sinusoidal or constant baselines and exponential, gamma, Weibull, or Lomax (power-law) excitation kernels, with AIC- and time-rescaling-residual-based kernel selection [3]; the full likelihood-derivation, asymptotic theory after Kwan–Chen–Dunsmuir, and Toronto Police Service application are developed in detail in the companion paper [9].

5.5 Psychometrics

The `psymet` module covers Cronbach's α, McDonald's ω, KMO sampling adequacy, Bartlett's sphericity, parallel analysis for factor retention, composite reliability, AVE, classical test theory item analysis, and item-response-theory estimation for one-, two-, and three-parameter logistic models (via `moirais.fn.irt1p`, `irt2p`, `irt3p`), with differential-item-functioning diagnostics (Mantel–Haenszel, logistic regression).

6 The MRM primary application

The MRM (McNamara-Ruhela-Medina) framework, developed in detail in the companion paper [6], is MOIRAIS's named primary application. It is a multi-source framework that brings five distinct Canadian sociolegal data streams under a coordinated set of estimators:

1. Provincial Ontario Tracking Information System (OTIS) restrictive-confinement microdata.
2. Federal Structured Intervention Unit Implementation Advisory Panel reports plus the four Sprott–Doob–Iftene academic replications.
3. Ontario Special Investigations Unit (police-oversight) director's-report corpus, ingested by an automated scraper bundled with MOIRAIS (§7).
4. Toronto Police Service open-data crime categories with the Statistics Canada Crime Severity Index.
5. Federal Corrections and Conditional Release Statistical Overview tables introduced into the public record by Doob's *T-539-20* affidavit.

MRM is implemented in MOIRAIS as a set of *MRM modules* spanning an aggregate-IRR family, a per-individual ten-estimator ensemble, a GEE clustering grid, a Doob χ^2 family for contingency tables, a Goffmanian institutional-churn analysis suite [4], and a Mandela Rules classifier [12] that operates at both the federal and provincial levels. The framework's headline empirical finding — that the provincial Ontario Mandela "torture"-classified proportion exceeds the federal SIU rate found by Sprott and Doob [11] — is documented in [6].

This paper does not duplicate the mathematical content of the MRM paper. The reader interested in the framework's theorems, estimator derivations, or empirical replication of Sprott–Doob–Iftene's published χ^2 statistics should consult [6].

7 Automated scrapers

MOIRAIS bundles two automated data-collection pipelines, both implemented as standalone modules with their own normalisation schemas:

7.1 Ontario SIU scraper

The `moirais.siu` package ingests the Ontario Special Investigations Unit director's-report corpus end-to-end:

$$\texttt{scrape\_drid} \to \texttt{parse\_html} \to \texttt{normalize} \to \texttt{schema} \to \texttt{write\_csv} \to \texttt{analyze}. \tag{1}$$

The output is a 65-column canonical CSV keyed on the SIU case number (YY-XXX-NNN). Seven `RichResult`-emitting analysers cover demographics, decision timing, mental-health-and-race indicators, by-police-service breakdowns, by-year breakdowns, and the case-count panel. The pipeline is conservative on the network side (drid ranges scraped politely with a concurrency limit and an HTTP cache), and all analytical outputs are reproducible from the canonical CSV. Note that this is the *Ontario police-oversight* SIU and is distinct from the federal Structured Intervention Unit (§6); I have deliberately chosen module names (`moirais.siu` for Ontario police-oversight, `moirais.siuiap` for the federal panel) that do not begin with a common prefix in order to make the distinction visible at import time.

7.2 Federal SIU IAP indexer

The `moirais.siuiap` module indexes (rather than scrapes) the federally-appointed SIU Implementation Advisory Panel reports, the four Sprott–Doob–Iftene CRIMSL / Schulich Law academic publications, and Doob's T-539-20 Federal Court affidavit. The module exposes citation strings, panel-member metadata, and pointers to the source PDFs; analyses that consume this content live in `moirais.sprott_doob` and `moirais.doob_trends`.

8 Polyglot REPL and language interop

The TUI's REPL screen supports a polyglot mode that auto-detects the language of each line (Python, R, Bash / Zsh / POSIX `sh`) and bridges variables bidirectionally across languages. Scalars, vectors, strings, booleans, and pandas `DataFrame`s travel across the bridge transparently. This is the recommended interface for users who need to mix Python's machine-learning ecosystem with R's statistical-modelling ecosystem in the same session. Language tags display as `[P]` (Python), `[R]` (R), `[Z]` / `[B]` / `[S]` (Z-shell / Bash / POSIX shell).

9 LLM integration

A multi-provider local-LLM chain provides assistive analysis without cloud dependencies:

1. Local Ollama (auto-detects available models, prefers a `perseus:e2b` expert model when present).
2. OllamaFreeAPI (free remote, no API key) via a vendored `moirais.fam` client (pure-`httpx`, no `pydantic` / Rust dependencies).
3. Gemini, via `GEMINI_API_KEY`.
4. Generic OpenAI-compatible, via `LLM_API_BASE_URL` and `LLM_API_KEY`.
5. A keyword-matched local fallback that runs offline.

The `moirais.quant` module implements a vendored TurboQuant [14] KV-cache compression that preserves the unbiased inner-product property critical for downstream causal-inference applications, with a C reference implementation (`quant_ggml.c`) bridged by ctypes. TurboQuant is data-oblivious (no calibration data) and unbiased, which is the property that lets us use it inside the causal- inference workflow without contaminating the estimators with the quantization error.

10 Emissions tracking

The `moirais.emissions` module is a pure-Python re- implementation of the codecarbon CSV format, covering 213 countries via Statistics Canada / IEA energy-mix data, US state-level EPA eGRID data, and Canadian provincial-grid data. The module avoids the `pydantic` / PyO3 dependency chain of the upstream codecarbon package, which means MOIRAIS emissions tracking works on Python 3.15+ where pydantic-core has not yet shipped wheels.

11 Reproducibility and continuous integration

- **Tests.** The Python test suite contains an extensive baseline test suite plus 55 tests of the federal/provincial Mandela cross-jurisdiction modules in the MRM primary, all passing under continuous integration.

- **Documentation.** A Sphinx documentation site builds without warnings under strict mode (`-W`), including auto-generated API reference for both the Python and R packages.
- **Distribution.** The package is released on PyPI (Python; Trusted Publisher OIDC, no API token), CRAN (R; multi-OS `R CMD check` matrix), and r-universe (R; nightly binary builds from GitHub source). Each GitHub release auto-mints a Zenodo software DOI via the GitHub-Zenodo integration with metadata pinned in `.zenodo.json`.
- **License.** MOIRAIS is GPL-2.0-only (Linus copyleft). The Ontario SIU scraper component (§7) is additionally subject to the Ontario open-government licence on the source corpus.

12 Limitations

Sociolegal scope. The MRM primary application is specific to Canadian sociolegal data streams. Adapting the framework to other jurisdictions requires user-supplied data and domain-specific operationalisations of the Mandela Rules classifier. I document this in [6].

Cryptography. The `crypto` module is intended as a teaching layer and as glue for low-stakes research, not a substitute for production cryptographic libraries. Users who need side-channel-resistant implementations or audited code should delegate to `cryptography` or `libsodium` bindings.

LLM dependencies. The local LLM chain depends on external services (Ollama daemon, OllamaFreeAPI community servers, or commercial Gemini / OpenAI APIs). The keyword-matched fallback keeps the package functional in fully offline / air-gapped environments, but it is not a substitute for a local LLM.

Test coverage. The 14k-test baseline does not yet include systematic property-based testing for the cryptographic primitives or fuzzing for the parser-heavy scrapers; both are on the roadmap.

Acknowledgements

The author thanks Prof. Beatrice Jauregui (University of Toronto) for the network of mentorship that made this line of work possible, and Prof. Ayobami Laniyonu (University of Toronto) for valuable lessons over the broader period of this work that informed the author's thinking on policing and corrections research. The MRM methodology lineage acknowledges the federal context provided by Anthony N. Doob's affidavit and the four Sprott–Doob–Iftene independent academic reports [11].

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